ADDING RADICALS

Simplify first, then combine

Simplifying first is what reveals a common factor when the radicals do not look alike at the start.

01

UNLIKE RADICALS — BEFORE

$$\sqrt{12} + \sqrt{27}$$

↓ Simplify each radical — √12 = 2√3 and √27 = 3√3

02

LIKE RADICALS

$$2\sqrt{3} + 3\sqrt{3}$$

↓ Combine the coefficients — (2 + 3)√3, just as 2x + 3x = 5x

03

SINGLE TERM — AFTER

$$5\sqrt{3}$$